Customer Churn Prediction: Business Summary Report

Prepared for: Management Team
Subject: Proactive Customer Retention and Churn Risk Analysis

1. Executive Summary
Customer churn (the rate at which customers cancel their subscriptions) directly impacts our recurring revenue and growth. Currently, our overall churn rate stands at approximately **26.5%**. Because acquiring a new customer is significantly more expensive than retaining an existing one, the goal of this initiative was to analyze historical customer data, identify why customers leave, and build a predictive system to flag at-risk customers *before* they cancel.

By leveraging a machine learning model, we can now score every active customer with a "Churn Probability." This empowers our retention team to transition from a reactive approach (trying to save a customer who is already canceling) to a **proactive approach** (targeting high-risk customers with specialized offers), ensuring our retention budget is spent where it saves the most revenue.

2. Key Drivers of Churn
Through extensive Exploratory Data Analysis (EDA) and model explainability (SHAP values), we identified the primary factors driving customer cancellations.

* **Contract Type is the Biggest Risk Factor:** Customers on **Month-to-month** contracts churn at a remarkably higher rate (over 42%) compared to those on One-year (11%) or Two-year (3%) contracts. Month-to-month customers lack long-term commitment and are highly sensitive to monthly billing cycles.
* **The "First Year" Danger Zone:** The highest volume of churn occurs within the first **0 to 12 months** of tenure. If a customer stays beyond their first year, their likelihood of churning drops significantly.
* **Fiber Optic Service:** Customers using our Fiber Optic internet service churn more frequently than those on DSL. This suggests a potential misalignment between the premium price of Fiber Optic and the perceived service quality or reliability.
* **Lack of Support Add-ons:** Customers who do *not* subscribe to supplementary services like **Tech Support, Online Security, and Device Protection** have a much higher churn rate. These add-ons create "stickiness" and make it harder for customers to switch providers.

3. The Predictive Model
We built and evaluated several machine learning algorithms (Logistic Regression, Random Forest, and XGBoost) to predict individual churn risk.

Rather than simply guessing "Yes" or "No", the model outputs a **Probability Score (0% to 100%)** for every customer.
* **High-Risk Segment (>75% probability):** Immediate intervention required.

* **Medium-Risk Segment (50% - 75% probability):** Monitor and send soft retention campaigns.
* **Low-Risk Segment (<50% probability):** Safe customers; do not offer unnecessary discounts.

Furthermore, we developed an **Interactive Dashboard** (Power BI) for executives to track global revenue at risk, and a **Streamlit Web Application** for retention agents to input a customer's profile and instantly see their churn risk and the specific reasons driving that risk.

4. Strategic Recommendations
Based on the data, we recommend implementing the following targeted strategies to immediately reduce churn and protect revenue:

1. **Incentivize Long-Term Contracts:**
   Since month-to-month customers are flight risks, run targeted campaigns offering a slight monthly discount or a free month of service if they upgrade to a 1-year or 2-year contract. The short-term discount will easily pay for itself through increased customer lifetime value.

2. **Revamp the Onboarding Experience:**
   With the first 12 months being the most critical, implement a "White-Glove Onboarding" program. Schedule proactive check-in calls at Month 3 and Month 6 to ensure new customers are satisfied and fully understand their services.

3. **Bundle "Sticky" Add-on Services:**
   Offer Tech Support and Online Security at a heavily subsidized rate for the first 6 months. Once customers integrate these security and support features into their daily lives, their likelihood of leaving drops drastically.

4. **Investigate Fiber Optic Quality:**
   Conduct a targeted survey specifically for Fiber Optic customers who recently churned. We must determine if the high churn rate is due to competitive pricing from rivals, or actual technical outages and speed issues.

5. Next Steps
By integrating this predictive model directly into our CRM, customer service representatives will see a "Churn Risk Score" on their screen when a customer calls in. Representatives can then be authorized to offer specific retention discounts *only* to those in the High-Risk segment, saving the company from offering blanket discounts to customers who were never planning to leave.

We recommend running a 30-day A/B test: apply these targeted retention strategies to half of the high-risk customers identified by the model, and compare their retention rate to the other half.